

# Tattoo & Street Art Visual Classifier

ITAI 1378  
MIDTERM  
BLUEPRINT  
PROPOSAL

A transfer-learning computer vision system categorizing body ink, outdoor murals, and background clutter.

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Project Tier

Tier 1: Core (single-Task CNN)

Institution

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# THE PROBLEM: IDENTIFYING DISTINGUISHING PHYSICAL MARKERS

## The Real-World Challenge

- In missing persons registries, disaster victim identification (DVI), and medical biometric records, being able to distinguish physical features such as tattoos, scars, and skin patterns is crucial for human identification.
- When searching through digital evidence, personal photos, or intake databases, automated image search tools constantly confuse outdoor murals, street graffiti, clothing prints, or plain skin with actual permanent body ink.

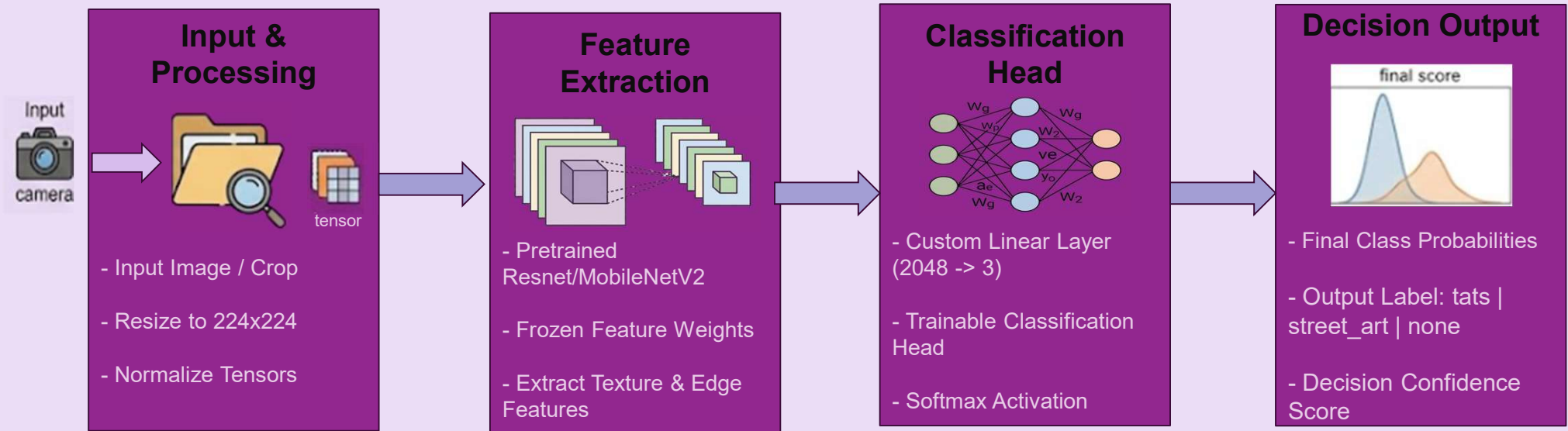
## Who Cares & Why Does It Matter?

- Who: Humanitarian recovery organizations, forensic pathologists, and digital registries.
- Why: Reviewing thousands of uncategorized personal photos manually to verify if an image depicts a real tattoo on a victim or missing person causes critical delays during recovery and identification efforts.

## What Will Solve It?

- An automated 3-class vision classifier (tats, street\_art, none) that filters out urban murals, graffiti, and background noise, isolating physical body markings for rapid review.

## THE SOLUTION: 3-STAGE TRIAGE PIPELINE



An end-to-end deep learning pipeline ingests raw field images, extracts texture and edge representations using a pretrained ResNet50 backbone, and classifies the input into tattoos, street murals, or background clutter to rapidly triage physical identification markers.

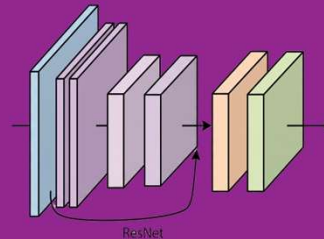
## TECHNICAL APPROACH & JUSTIFICATION

### CV Technique



- **Multi-Class Image Classification**
- **3 target classes:** tats, street\_art, none
- **Focus:** Automated visual triage isolating human skin ink from outdoor street artwork

### Model Architecture



- **ResNet50 Backbone** (ImageNet Pretrained)
- **Transfer Learning:** Frozen feature extraction layers to preserve generalized weights
- **Custom Head:** Linear classification layer ( $2048 \rightarrow 3$ ) + Softmax

### Framework & Tools

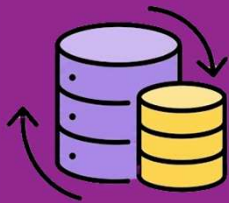


- **PyTorch & Torchvision** for neural network modeling and transforms
- **Data Loaders:** Standard ImageFolder directory structure
- **Hardware:** Google Colab with free-tier T4 GPU acceleration

**Why This Approach:** Transfer learning via ResNet50 is optimal for a 2,600-image dataset. Pretrained convolutional layers extract rich edge and texture features out of the box, preventing severe overfitting while allowing the model to train efficiently within Colab's GPU session limits.

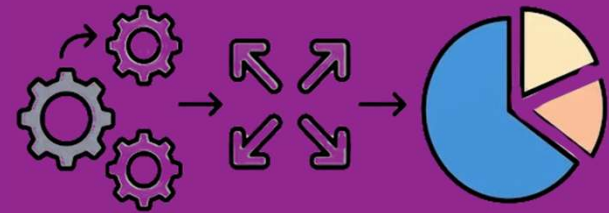
## DATA PLAN & ARCHITECTURE

### Dataset Composition & Class Distribution



- **Total Volume:** 2,661 Images across 3 classes
- **tats:** 802 images (skin ink, line work, pigmentation)
- **street\_art:** 695 images (murals, spray paint on urban surfaces)
- **none:** 1,164 images (billboards, traffic signs, clear and damaged skin, noise)
- **Sources:** Curated from Roboflow Universe and Images taken by Me.

### Data Pipeline & Split Strategy



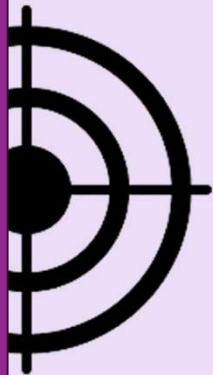
- **Train Split (80%):** 2,128 (931 none, 641 tats, 556 street\_art)
- **Validation Split (10%):** 265 (116 none, 80 tats, 69 street\_art)
- **Test Split (10%):** 268 (117 none, 81 tats, 70 street\_art)
- **Resolution & Tensors:** Resized to 224x224 RGB, ImageNet Normalization
- **Augmentations:** Random Horizontal flip, slight rotation, color jitter

**Data Labeling Challenge & Mitigation:** Fine tattoo linework can visually mimic intricate urban murals on building surfaces (Graffiti). Mitigated by heavily weighting the *none* class (1,164 images, including bare/damaged skin, street signs, billboards, and blank walls) to force the model to learn localized dermis context rather than just raw edges or pigment.

# EVALUATION CRITERIA & SUCCESS METRICS

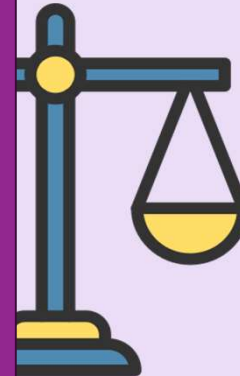
## Primary Performance

- **Target Metric:** 80% Overall Multi-Class Top-1 Accuracy
- **Operational Goal:** Establish reliable automated routing across body ink, street murals, and negative background clutter.
- **Optimization Loss:** Cross-Entropy Loss with Softmax activation on final logits.
- **Minimum Threshold:** Model must sustain > 75% accuracy across all three individual classes to prevent one class from dragging down the aggregate score.



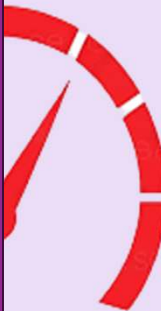
## Secondary Evaluation

- **Balance Metric:** Macro F1-Score  $\geq 0.78$  to account for class imbalance (1,164 *none* vs. 695 *street\_art*).
- **Priority Recall:** Tattoo Recall target  $\geq 85\%$  to ensure actionable identification markers are rarely missed in intake workflows.
- **Diagnostic Audit:** Per-class Precision/Recall Analysis and a 3x3 Confusion Matrix to pinpoint whether false positives stem from skin textures or wall patterns.



## Baseline & Resource Constraints

- **Chance Baseline:** 33.3% (random assignment across 3 classes).
- **Zero-Rule Majority Baseline:** 43.7% (predicting *none* on all inputs: 1,164/2,661).
- **Inference Budget:** < 200 ms per image on standard CPU instances; < 25ms on CUDA-enabled runtimes.
- **Storage Footprint:** Quantized model weights kept at < 100MB for lightweight containerization.



**Model Selection Protocol:** Checkpoints will be ranked by the lowest Validation Loss rather than accuracy alone. Training terminates early if validation loss does not improve after 5 epochs to prevent memorization of repetitive skin tones or background noise. Final verification occurs solely on the untouched 268-image test set.

## PROJECT MILESTONE PLAN & TIMELINE

| Phase                                                                                                 | Project Goal                                                                                       | Milestone                               | 16 Week Term |
|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------|--------------|
|  Blueprint           | Project Proposal & architecture plan approved                                                      | Midterm Pitch Submitted                 | Week 10      |
|  First Working Demo  | Pretrained ResNet50 pipeline runs end-to-end on sample batches                                     | Pipeline verifies inference end-to-end  | Week 11      |
|  Make it Yours       | Ingest the full 2,661 dataset; fine-tune the head on <i>tats</i> , <i>street_art</i> , <i>none</i> | Model actively trains on custom classes | Weeks 12-13  |
|  Improve & Measure   | Measure against targets ( $\geq 80\%$ Acc, $\geq 0.78$ F1); tune augmentations.                    | Test metrics & confusion matrix logged  | Week 14      |
|  Package & Present | Finalize PyTorch weights (100MB), README, demo video & deck                                        | Final deliverables submitted            | Week 15      |

**Why Does the Demo Precede Full Data Work?** In industry workflows, proving the ResNet50 inference pipeline runs end-to-end on a small sample de-risks deployment early. This confirms code and GPU environments work before investing compute time in training across all 2,661 images.

## RISKS & COMPUTATIONAL RESOURCES

### Domain Overfitting



- **Risk:** High visual similarity between intricate tattoo linework and fine mural line art leads to misclassifications.
- **Impact:** False positives where human skin with tattoos is flagged as street graffiti or vice versa.
- **Plan B:** Unfreeze deeper ResNet50 residual layers to learn fine-grained substrate textures and apply heavy color jitter + affine augmentation.

### Class Imbalance Risk



- **Risk:** Skewed training set (1,164 *none* vs. 695 *street\_art*) can cause the model to default toward predicting the majority class.
- **Impact:** High overall accuracy but poor recall on street murals and tattoos.
- **Plan B:** Implement class-weighted cross-entropy loss to penalize minority-class errors, or apply oversampling via augmentations on *street\_art*.

### Compute & Estimated Cost



- **Primary Platform:** Google Colab with free-tier NVIDIA T4 GPU (16GB VRAM).
- **Backup Compute:** Kaggle Notebooks (free P100/T4 GPUs, 30hrs/week).
- **Local Fallback:** CPU workstation for evaluation and batch testing
- **Estimated Cost:** \$0.00 (fully leveraging accessible, free-tier cloud GPU resources).

**Risk Protocol:** Model checkpoints are monitored against per-class F1-scores during training. If any single class drops below 70% recall, Plan B interventions (class weighting and deep layer unfreezing) are triggered before the final test evaluation.